

AFRICA MEDICAL COLLEGE

Final Examination in Histology and Embryology for Year I Medical Students

Time allowed: 3:00 hours.

Date: Feb 18, 2013

Name: _____

I. HISTOLOGY

For each of the following Multiple Choice Questions choose the one BEST appropriate answer and mark 'X' in the appropriate box located against the corresponding question number and letter in the Separate Answer Sheet provided.

_____1. **1000 micrometers** is equivalent to:

- a) 1 centimeter. b) 100 millimeters. c) 10 millimeters. d) 1 millimeter. e) 0.1 millimeter

_____2. The stain of choice to demonstrate **glycogen** and/or **mucus** in cells is:

- a. hematoxylin. b. eosin. c. lipofuchsin. d. hemosiderin. e. periodic acid Schiff (PAS).

_____3. With regard to the **cellular organelles**, which statement is **true**?

- a) The nucleolus is the site of the assembly of ribosomal subunits.
b) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion.
c) Lysosomal enzymes have optimal activities at basic pH and they do synthesize rRNA and DNA.
d) Golgi apparatus possesses in its cis-face heterochromatin which is transcriptionally active.
e) Steroid hormones are synthesized by peroxisomes for they contain shelf-like mitochondria.

_____4. Concerning the **morphology of a cell**, select the **correct statement**.

- a) Microtubules are more stable than microfilaments.
b) G₀ phase is the longest and most variable cell-cycle phase in continuously dividing cells.
c) Ion-transporting cells contain abundant basal mitochondria between infoldings of the basal plasma membrane.
d) Ribosomes are membranebounded organelles and are involved in the breakdown of hydrogen peroxide.
e) Centrioles are associated with membrane activities such as endocytosis, exocytosis, and cell migratory activities.

_____5. With regard to the **epithelial tissue**, select the **incorrect statement**.

- a) The lining of the urinary bladder is transitional epithelium.
b) Holocrine secretion is the typical mode of secretion of DNES and APUD cells.
c) An example of epithelioid tissue is the parenchyma of the adrenal [suprarenal] gland.
d) Endothelium is another name for simple squamous epithelium that lines the vascular system.

e) Cilia are epithelial cell surface [apical] specializations used to move mucous and particles along a surface.

———6. With regard to the **epithelial tissue**, which statement is **true**?

- a) Conjunctiva is lined by stratified columnar epithelium.
- b) Desmosomes permit the passage of ions between adjacent cells.
- c) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.
- d) Arteriovenous anastomosis provides nutrition to the highly cohesive and polarized rows of cells.
- e) The major constituents of epithelia is abundant extracellular matrix with amorphous ground substance.

———7. **Mesothelium** lines:

- a. the respiratory and the urinary tracts.
- b. the gastrointestinal and reproductive tracts.
- c. the nasal and the oral cavities.
- d. the peritoneal, pleural and pericardial cavities.
- e. the central nervous system and the muscular system

———8. Which surface is **NOT lined by epithelial tissue**?

- a. surface of skin.
- b. cornea surface of eye.
- c. luminal surface of intestinal tract.
- d. luminal surface of trachea and alveoli of lungs.
- e. surfaces of bones and muscles.

———9. With regard to the **connective tissue**, which statement is **false**?

- a) Mucous connective tissue lacks connective tissue cells.
- b) Collagen is characterized by high tensile strength and flexibility.
- c) Plasma cells characteristically synthesize and secrete immunoglobulins.
- d) In ordinary connective tissue, the predominant component of ground substance is water.
- e) Fibroblast produces and maintains all extracellular matrix components of connective tissue.

———10. Which **connective tissue cell type** is functionally related to the **basophils**?

- a) Fibroblast
- b) Mast cell
- c) Histiocyte
- d) Plasma cell
- e) Myofibroblast

———11. With regard to **cartilage**, which statement is **false**?

- a) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- b) Hyaline cartilage is characterized by an amorphous glassy matrix.
- c) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- d) Fibrocartilage is present in regions subjected to high tension.
- e) Fibrocartilage is used for endochondral bone ossification.

———12. **Elastic cartilage**,

- a) is found in the annulus fibrosus and intervertebral disks.
- b) the collagen in its matrix is mainly type II.
- c) is the primary skeletal tissue in the fetus.
- d) has no identifiable perichondrium.
- e) is the most widely distributed cartilage type in the body.

———13. With regard to **bone**, which statement is **false**?

- a) Compact (dense) bone is characterized by the presence of osteons.
- b) Bone formation without a pre-existing cartilage model is called intramembranous bone formation.
- c) In the restructuring of the skeleton, bone remodeling follows a cycle in which bone formation alternates with resorption.
- d) Osteoblasts secrete the components of bone matrix.
- e) Osteocytes are involved in bone resorption.

———14. Which of the following statements is **correct** regarding the **bone**?

- a) Primary bone has a higher mineral content than the secondary bone.

- b) Sharpey's fibers are calcified collagen bundles that run in concentric manner.
 c) Bone contains matrix with more inorganic components than organic component in its dry weight.
 d) Osteoclasts have processes that run in their canaliculi. e) Osteoblast is derived from blood monocytes.
- 15. Which of the following cells **does not** belong to the **mononucleated phagocytic system**?
 a) Micoglia b) Dust cells c) Langerhan's cells d) Kupffer cells e) Hepatocytes
- 16. With regard to the **muscular system**, which statement is **false**?
 a) Z-lines represent attachment sites for thick filaments (myosin).
 b) Cardiac muscle consists of individual cells attached end-to-end by intercalated disks.
 c) The distance between two Z-lines in skeletal muscle is equivalent to the length of a sarcomere.
 d) The electrical signal of an action potential is conveyed into the interior of each muscle fiber by transverse tubules.
 e) Skeletal muscle can be described as syncytial, with many nuclei per "cell" located at the periphery of its muscle fibers, close to the fiber membrane.
- 17. The sense organ most closely associated with **stretch reflexes** (e.g., knee jerk) is the:
 a. Meissner's corpuscle. b. Pacinian corpuscle. c. Muscle spindle.
 d. Golgi tendon organ. e. Free nerve ending.
- 18. Which of the following cell types is found predominantly in **white matter**?
 a) fibrous astrocytes b) microglia c) satellite cells d) protoplasmic astocytes e) Schwann cells
- 19. With regard to the **dendrites**, which of the following statements is **true**?
 a) each has a constant diameter along its entire length. b) may contain Nissl bodies.
 c) contain neurosecretory vesicles. d) are found in white matter. e) are typically myelinated.
- 20. Which of the following statements is **true** concerning the **nervous tissue**?
 a) Schwann cells provide myelin for several axons simultaneously.
 b) Action potentials normally travel from the axon terminal toward the cell body.
 c) Unmyelinated axons generally have a smaller diameter than myelinated axons.
 d) Gray matter in the central nervous system consists of axon tracts with abundant myelin.
 e) Microglia line the central canal of the spinal cord and the ventricular system of the brain.
- 21. A typical **peripheral mixed nerve** includes all of the following **EXCEPT**:
 a) Schwann cells. b) astrocytes. c) sensory axons d) motor axons
 e) connective tissue of the perineurium and/or endoneurium
- 22. Which of the following statements is **false** about the **cerebrospinal fluid**?
 a) Cerebrospinal fluid is actively secreted by ependymal cells of the choroid plexus.
 b) An imbalance between production and drainage of cerebrospinal fluid can cause hydrocephalus.
 c) Compared to blood, normal cerebrospinal fluid contains much higher levels of potassium and calcium.
 d) Arachnoid mater forms trabeculae through which cerebrospinal fluid flows. e) None of the above.
- 23. Which of the following statements is **true** regarding the **nervous system**?
 a) Choroid plexus is found in the central canal of the spinal cord.
 b) Pyramidal cells are characteristic of the ventral horn of spinal cord.
 c) Purkinje cells are characteristic of the cerebral cortex. d) White matter consists mainly of axons and neuroglial cells.
 e) Ependymal cells helps regulate the interstitial environment of central nervous tissue, serving as a buffer

between nerve cells and capillaries.

———24. **Dura mater** consists of:

- a. loose connective tissue with CSF as ground substance.
- b. dense fibrous connective tissue.
- c. white matter.
- d. gray matter.
- e. a thin layer of cells and collagen adhering to outer surface of the brain.

———25. Which of the following statements is **not true** regarding the **supporting cells in the nervous tissue**:

- a) Microglia are the smallest in size among the neuroglial cells.
- b) Astrocytes serve in conducting metabolites between blood vessels and neurons.
- c) Satellite cells surround the perikarya of the spinal ganglia in a stratified cuboidal fashion.
- d) Oligodendrocytes are found in both gray and white matters of the central nervous system.
- e) Ependymal cells line the central canal of the spinal cord and the ventricular system of the brain.

———26. Which of the following statements is **false** with regard to the **circulatory system**?

- a) Lymphatic capillaries can penetrate the media of veins.
- b) Vasa vasorum are more frequent in veins than in arteries.
- c) The tunica adventitia tends to be much larger in veins than in arteries.
- d) The concentration of glycosaminoglycans is higher in arterial than in venous tissue.
- e) Sinusoids of the liver and the glomeruli of the kidney typically possess continuous endothelium.

———27. With regard to the **white blood cells**, which of the following statements is **true**?

- a) Neutrophils are (normally) the most common white blood cells in circulating blood.
- b) A patient with a parasitic infection is likely to reveal an increase in the circulating monocytes.
- c) Basophils normally re-enter the bloodstream after leaving circulation and immigrating into peripheral tissues.
- d) A macrophage can originate from a mast cell.
- e) None of the above.

———28. The inner layer of a **blood vessel wall**, characterised by a **simple squamous endothelium** supported by a thin layer of connective tissue, is the tunica:

- a. adventitia.
- b. external elastic lamina.
- c. media.
- d. internal elastic lamina.
- e. intima.

———29. Which of the following statements is **false** with regard to the **circulatory system**?

- a. Post-capillary venules lack a smooth muscle media.
- b. Carotid bodies are chemoreceptors sensitive to hypoxia.
- c. The brachiocephalic and subclavian arteries are elastic arteries.
- d. Continuous capillaries often have phagocytic cells inserted between the endothelial cells of their lining.
- e. A prominent inner elastic membrane (internal elastic lamina), often appearing in cross section as a wavy line, is characteristic of muscular arteries..

———30. Which of the following statements is **true** with regard to the **circulatory system**?

- a. Albumin is present in higher concentrations in plasma than in serum.
- b. Smooth muscle is most substantial in the media of muscular arteries.
- c. Heart valves normally consist of an endothelial surface covering loose areolar connective tissue.
- d. Pericytes, which are terminally differentiated and multinucleated, are specialized cardiac muscle cells.
- e. The SA node, the AV node, and the Purkinje fibers of the myocardium all consist of specialized nerve cells.

———31. Which of the following statements is **false** with regard to the **histology of the heart**?

- a. Ventricles contain more abundant elastic fibers than do the atria.
- b. Adipose tissue that generally surrounds the heart accumulates in the epicardium.

- c. The coronary arteries branch to supply the epicardium, myocardium, and endocardium,
- d. Endocardium is the thickest in atria and the thinnest in ventricles, especially the left ventricle.
- e. Cardiac muscle is more vascularized and has more abundant mitochondria than does skeletal muscle.

—32. Which of the following is partly **encapsulated and covered by nonkeratinized stratified squamous epithelium**?

- a) palatine tonsil
- b) lingual tonsil
- c) appendix
- d) Peyer's patch
- e) pharyngeal tonsil

—33. Which of the following **lymphoid organs** receives **afferent lymphatic vessels**?

- a) appendix
- b) palatine tonsil
- c) spleen
- d) thymus
- e) lymph node

—34. **Thymus-dependent** region of a peripheral lymphoid organs include,

- a) medullary cords of lymph nodes.
- b) periarterial lymphatic sheaths of the spleen.
- c) Hassall's corpuscles of the thymus.
- d) deep crypts of the pharyngeal tonsils.
- e) red pulp cords (Billroth's cords) of the spleen.

—35. Regarding the **thymus**:

- a) Has lymphoid nodules in its cortex.
- b) Is relatively more developed with increasing age.
- c) among blood vessels, contains only sinusoids.
- d) Has blood-thymus barrier in the cortex.
- e) Contains more B lymphocytes than the T lymphocytes.

—36. The **spleen**:

- a) Has smooth myocytes in its capsule.
- b) Filters lymph.
- c) Has no B lymphocytes.
- d) Has an outer cortex and inner medulla.
- e) Contains periarterial lymphoid sheath in its red pulp.

—37. Which of the following components of the **respiratory tract** wall increase(s) in amount from trachea to alveoli?

- a) cilia
- b) smooth muscle
- c) elastic fibers
- d) cartilage
- e) goblet cells

—38. Which of the following structures is (are) partly lined by **nonkeratinized stratified squamous epithelium**?

- a) nasopharynx
- b) larynx
- c) bronchi
- d) trachea
- e) bronchioles

—39. In which of the following are **goblet cells** most frequent?

- a. trachea
- b. bronchus
- c. bronchiole
- d. alveolus
- e. The frequency of goblet cells in each of the above is about equal.

—40. As **respiratory passages branch from trachea to bronchioles**, the epithelium gradually changes from:

- a. simple squamous to ciliated, pseudostratified columnar.
- b. simple columnar to simple squamous.
- c. simple cuboidal to simple columnar.
- d. stratified squamous to simple squamous.
- e. ciliated, pseudostratified columnar to simple cuboidal.

—41. **Alveolar type I cells** are:

- a. cuboidal cells that secrete surfactant.
- b. squamous cells involved in gas exchange.
- c. ciliated cells that move mucus.
- d. columnar cells that secrete proteolytic enzymes.
- e. ameboid and phagocytic cells that clean the alveolar surface.

—42. **Bronchioles**:

- a) Have well developed hyaline and elastic cartilages.
- b) Are lined by stratified squamous epithelium.
- c) Have no muscle.
- d) Have Clara cells.
- e) Have mucous producing glands.

———43. **Pulmonary surfactant**,

- a) prevents alveolar collapse by increasing surface tension.
- b) consists of a monolayer of phospholipids overlying an aqueous hypophase.
- c) excessive production in the lungs of premature infants causes respiratory distress syndrome.
- d) is secreted by type I pneumocytes.
- e) may promote the growth of bacteria.

———44. **Nasal mucosa** includes:

- a. pseudostratified, ciliated columnar epithelium.
- b. lamina propria.
- c. seromucous glands.
- d. venous plexuses.
- e. all of the above.

———45. **Cells of epidermis** that form the **stratified squamous keratinized epithelium** are:

- a) Keratinocytes
- b) Melanocytes
- c) Langerhan's cells
- d) Merkel's cells
- e) Chondrocyte

———46. Which of the following **epidermal layers** is most likely to contain both **desmosomes & hemidesmosomes**?

- a) stratum basale
- b) stratum corneum
- c) stratum granulosum
- d) stratum lucidum
- e) stratum spinosum

———47. Which of the following is (are) often **absent in thick skin** but typically present in **thin skin**?

- a) arrector pili muscles
- b) Meissner's corpuscles
- c) sweat glands
- d) stratum basale
- e) stratum spinosum

———48. **Arrector pili muscles**,

- a) are composed of bundles of myoepithelial cells
- b) have no effect on hair position in humans.
- c) help push sebum onto the skin surface.
- d) contain capillary networks that nourish growing hairs.
- e) attach directly to the external root sheath by desmosomes.

———49. The **glands** most closely associated with **hair follicles** are:

- a. eccrine sweat glands.
- b. apocrine sweat glands.
- c. sebaceous glands.
- d. lymphatic glands.
- e. submucosal glands.

———50. Which of the skin's many functions are partly or completely provided by components of the **dermis**?

- a) protection from dehydration
- b) regulation of blood pressure
- c) synthesis of keratin
- d) protection from ultraviolet radiation
- e) synthesis of melanin

———51. Which of the following statements is **true** about the **lip**?

- a) The inner surface is lined by stratified columnar epithelium.
- b) The inner aspect has sebaceous glands.
- c) The outer aspect has apocrine sweat glands.
- d) The transitional border has abundant salivary glands.
- e) The core contains skeletal muscles.

———52. Which of the following statements is **true** regarding the **teeth**?

- a) Enamel is found overlying dentin in both crown and root portion of the teeth.
- b) Odontoblastic processes branch and taper as they approach dentinoenamel junction.
- c) Enamel develops from mesoderm, while both dentin and cementum develop from overlying ectoderm.
- d) Dentin is not present in the deciduous teeth.
- e) Enamel has more collagen fibres than dentin.

———53. **Submucosal glands** occur in the:

a. Duodenum and esophagus.

b. Stomach and ileum.

c. Jejunum and colon.

d. Appendix and anal canal.

e. All of them.

———54. Which of the following regions of the **GI tract** is characterized by **stratified squamous epithelium**?

a. Duodenum

b. Jejunum

c. Ileum

d. Colon

e. Anal canal

———55. **Permanent circular folds (plicae circulares)** are characteristic structural features of the walls of which of the following organs?

a) Anal canal

b) Colon

c) Small intestine

d) Cecum

e) Stomach

———56. Regarding the **stomach**:

a) Has its muscularis externa made by skeletal muscle fibres.

b) Characteristically contains the most goblet cells in the mucosa covering its villi.

c) Mucosal glands of the pylorus contain abundant parietal cells in their basal half.

d) Has basophilic oxyphil cells in its glands that secrete HCl.

e) None of the above.

———57. Which of the following statements is **correct** regarding the **intestine**?

a) Intestinal glands in the large intestine are branched alveolar type.

b) The striated border of the small intestine is made by microvilli.

c) Small intestine has taenia coli.

d) Large intestine has villi.

e) Both the small and the large intestines are able to withstand the acidic environment because they are lined with stratified squamous parakeratinized epithelium.

———58. Which one of the following marks best **ileum** from **duodenum and jejunum**?

a) Long and finger like villi.

b) Well developed Peyer's patches.

c) Well developed plicae circularis.

d) Well developed microvilli.

e) Simple columnar lining epithelium.

———59. **Vermiform appendix**,

a) is a blind-ended evagination of the ileum.

b) has few lymphoid nodules in its walls.

d) contains short villi.

c) has relatively shallow intestinal glands.

e) longitudinal smooth muscle gathers into its teniae coli.

———60. Like the small intestine, the **gall bladder**:

a. is lined by columnar cells with a brush border of microvilli.

b. has goblet cells in the mucosal epithelium.

c. has crypts with Paneth cells at the bottom.

d. has villi with lacteals in the core.

e. Myoepithelial cells are responsible for emptying the gallbladder.

———61. Which of the following is **correct** about the **histology of pancreas**?

a) The exocrine cells are arranged as simple tubular type.

b) The exocrine portions have more capillaries than the endocrine portion.

c) The islets of Langerhans are made by larger and more eosinophilic cells than the exocrine cells.

d) Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.

e) The B cells of islets of Langerhan's are distinguished under light microscope by being larger and more basophilic than the other cells of the islets.

———62. **Kupffer cells**,

a) are nonphagocytic

b) contain lysosomes

c) are derived from circulating neutrophils

d) resemble hepatocytes in HE-stained sections

e) border on the bile canaliculi

———63. **Hepatocyte** function includes all of the following **EXCEPT**:

a. drug detoxification.

b. protein synthesis.

c. breakdown of hemoglobin.

d. gluconeogenesis.

e. urea formation by deamination of amino acids.

———64. Select the **incorrect** statement with regard to the **GI tract**.

- a. The filiform papillae, which lack taste buds, consist of whitish tips made up of keratinized epithelial tissue.
- b. The basal surface of hepatocytes is bathed by blood plasma as it flows freely through the space of Disse.
- c. The principal secretory product for the Paneth cells clustered at the ends of intestinal crypts is lysosomal enzymes.
- d. Brunner's glands empty their secretions into the lacteals.
- e. Folds in the mucosa of the stomach are called rugae.

———65. What vessels arise from the **efferent arterioles**?

- a. Afferent arteriole
- b. Efferent arteriole
- c. Capillary tuft
- d. Vasa recta
- e. Peritubular capillaries

———66. Select the **correct statement** pertaining to the **morphology of the kidney**.

- a. Macula densa cells are located in the urinary pole of the nephrons.
- b. The distal convoluted tubules exhibit brush borders and apical canaliculi.
- c. The renal pyramid and its associated cortex is referred to as the renal lobe.
- d. Interstitial cells in the renal medulla, called mesangial cells, secrete erythropoietin through filtration slits.
- e. Renin is secreted by the podocytes from the parietal layer of Bowman's capsule referred to as area cribrosa.

———67. In the **urinary system**, which statement is **true**?

- a. The male urethra contains transitional epithelium throughout its length.
- b. Aldosterone production is one of the most important function of the kidney.
- c. Endothelial cells of glomerular capillaries are of fenestrated variety without a diaphragm.
- d. Distal convoluted tubule is longer and more convoluted than proximal convoluted tubule.
- e. Mesangial cells play an important role in the maintenance of acid-base balance in the blood.

———68. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?

- a) ADH
- b) ACTH
- c) FSH
- d) TRH
- e) TSH

———69. Which statement is **true** with regard to the **endocrine system**?

- a) Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in bones.
- b) The secretory activity of the epiphysis cerebri {pineal body} is stimulated by darkness and inhibited by light.
- c) The chromophobes of pars distalis of adenohypophysis maintain favorable fluid and electrolyte environment for the optimal activities for the chromophil secretory cells.
- d) The basophilic oxyphil cells are the predominant cell type in the parathyroid gland which inhibit the activity of osteoclasts by decreasing blood calcium levels .
- e) The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained with potassium bichromate.

———70. The **neurohypophysis**,

- a) contains chromophils.
- b) includes the pars intermedia.
- c) secretes ACTH.
- d) contains axons of hypothalamic neurons.
- e) hormone production is regulated by releasing factors.

———71. In the **endocrine system**, which statement is **false**?

- a. The base of the hypothalamus which connects with the pituitary is called the median eminence and infundibulum.
- b. Pancreatic alpha cells secrete glucagon.
- c. Estrogen and progesterone are secreted by mammary glands.
- d. Adrenal medulla is most similar to a sympathetic ganglion.
- e. None of the above.

———72. **Testis** has the following coverings **EXCEPT**:

- a. tunica vasculosa b. tunica albuginea c. tunica adventitia d. tunica vaginalis e. None

———73. In the **testis**, which statement is **true**?

- a. The seminiferous tubules are lined with a complex stratified epithelium.
b. Interstitial cells (of Leydig) produce testicular fluid which constitute the bulk of ejaculate volume.
c. The secondary spermatocytes are the largest and long-lived diploid cells of the spermatogenetic lineage
d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis and rete testis.
e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

———74. Which of the following statements is **false** in the **male reproductive system**?

- a. The penile urethra passes through the corpus spongiosum.
b. The secretion of seminal vesicle is particularly rich in fructose.
c. The prostate gland is composed of compound tubuloalveolar glands.
d. The ejaculatory duct connects the seminal vesicle with the prostatic urethra.
e. Tunica albuginea, which is lined with pseudostratified columnar epithelium, is the thickest in epididymis.

———75. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains large mucosal folds that are extensively branched.
b) Contains ciliated epithelial cells. c) Contains mucus-secreting peg cells.
d) Contains smooth muscle in its walls. e) Is covered by serosa derived from the peritoneum.

———76. Which of the following statements is **true** about the **female reproductive system**?

- a. The life span of corpus luteum of pregnancy is about 10 lunar months.
b. Primary oocyte completes its first meiotic division at the time of fertilization.
c. Corpora albicans are ovarian follicles which degenerate after failing to mature.
d. Fornix of vagina is lined by stratified squamous epithelium. e. Mucosa of the uterus is commonly called perimetrium.

———77. Which of the following statements is **true** in the **female reproductive system**?

- a. Germinal (surface) epithelium covering the ovary gives rise to ova.
b. Estrogen and progesterone are secreted by the gonadotropic cells of the adenohypophysis.
c. The proliferative phase of the endometrium is characterized by a stratified squamous surface.
d. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.
e. After ovulation, granulosa cells and cells of theca interna grow and differentiate to form the corpus luteum.

———78. Which of the following statements is **false** about the **sense organs**?

- a. There are more pain receptors than cold receptors in the body.
b. Fingertips have more touch receptors than the skin of the back.
c. End bulbs of Krause possess receptors for cold and pressure reception.
d. Pacinian corpuscles are large, ellipsoid encapsulated, rapidly-adapting receptors.
e. Meissner's corpuscles are unencapsulated, myelinated receptor terminals located in the deep dermis and hypodermis.

———79. The **cornea**,

- a) is avascular. b) constitutes the anterior half of the tunica fibrosa. c) is composed of 10 layers.
d) accounts for eye color. e) is attached to the ciliary body by zonules.

———80. With regard to the **eye**, one of the following statements is **true**.

- a) Aqueous humor is actively secreted by cells of the stria vascularis.
b) Inner nuclear layer of the retina contains nuclei of photoreceptor cells.
c) Lacrimal canaliculi are lined by a stratified ciliated columnar epithelium.
d) An imbalance between production and drainage of aqueous humor can cause glaucoma.

e) The cornea is covered by an incomplete layer of pigmented cells and fibroblasts on its anterior surface.

———81. In the **ear**, one of the following statements is **false**.

- a) Otoliths are normally found in the saccule and utricle.
- b) Endolymphatic sac contains phagocytic cells like macrophages and neutrophils.
- c) The scala vestibule and tympani communicate with each other at the helicotrema.
- d) Tectorial membrane is secreted by the mediolus of the ampullae of the semicircular canals.
- e) The structure of the internal ear that contains special auditory receptors is the organ of Corti.

———82. With regard to the **ear**, one of the following statements is **true**.

- a) The bony labyrinth is filled with endolymph.
- b) The round window separates the middle ear from the eustachian tube.
- c) Reissner's membrane separates the scala media from the oval window.
- d) The middle ear contains modiolus, cristae ampullaris and semicircular canals.
- e) Hair cells of the ear, which are columnar mechanoreceptor cells, are bipolar neurons of the spiral ganglion.

———83. Which type of **neurones of the cerebral cortex** are predominantly found in the **primary sensory cortex**?

- a) Cells of Martinotti
- b) Horizontal cells of Cajal
- c) Fusiform cells
- d) Stellate cells
- e) Pyramidal cells

———84. The correct order of the three **layers of cerebellar cortex**, outer to inner is:

- a) Molecular layer, Purkinje cell layer and granular layer.
- b) Granular layer, Purkinje cell layer and molecular layer.
- c) Purkinje cell layer, molecular layer and granular layer.
- d) Granular layer, molecular layer and Purkinje cell layer.
- e) Molecular layer, granular layer and Purkinje cell layer.

———85. Which of the following statements is **not true** regarding the **meninges**?

- a) Arachnoid mater is delicate and transparent impermeable membrane.
- b) Dura mater of the spinal cord is made up of only one layer of dense fibrous membrane.
- c) Cranial dura mater has an outer layer which is connected by Sharpey's fibres to the calvaria.
- d) Arachnoid granulations are sites of formation and drainage of cerebrospinal fluid into the brain ventricles.
- e) Arachnoid mater has numerous trabeculae which are connected to the dura mater traversing the subdural space.

———86. **Regions of cell bodies in the central nervous system** make all of the following, **EXCEPT**:

- a) Gray matter
- b) Column
- c) Layer
- d) Nuclei
- e) None of the above

———87. Where would you localize the **gray matter in the central nervous system**?

- a. Outer surface of the cerebrum and cerebellum.
- b. Outer surface of the spinal cord.
- c. Inner portion of the spinal cord.
- d. Both a and b.
- e. Both a and c.

———88. Which of the following statements is **false** in the **central nervous system**?

- a) Both motor neurons and interneurons are pseudounipolar.
- b) A ganglion is a collection of neuron cell bodies outside of the central nervous system.
- c) The conglomerations of gray matter deep within the cerebrum and cerebellum are called nuclei.
- d) Basket cells are found in the cerebellum.
- e) The vast majority of neurons are interneurons.

_____89. Which of the following is an **element of the central nervous system**?

- a. Receptors b. Brachial plexus c. Sciatic nerve d. Ganglia e. Spinal cord

_____90. All of the following are **properties of neurons, EXCEPT**:

- a. High metabolic rate b. Extremely long longevity, lasting a lifetime. c. Specialized for conduction
d. High mitotic rate e. Limited ability for oxygen deprivation

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II. EMBRYOLOGY

_____1. Concerning **gametogenesis**, which statement is **true**?

- a) All oogonia become primary oocytes during ovulation.
b) Primary oocytes undergo cytodifferentiation to form three polar bodies prior to ovulation.
c) Secondary oocyte completes its second meiotic division to become a mature ovum at fertilization.
d) The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
e) Primary spermatocytes begin the first meiotic division before birth, but completion of prophase does not occur until adolescence.

_____2. Which statement is **true** about **female reproductive system**?

- a) The usual site of fertilization is the ampulla of the uterine tube.
b) If an oocyte is fertilized, the corpus luteum involutes and degenerates.
c) The endometrium has a wall made up of a very thick smooth muscle layer.
d) The perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
e) The vaginal glands are sloughed off and discarded with each menstrual flow during pregnancy.

_____3. With respect to the **zona pellucida**, which statement is **false**? The **zona pellucida**:

- a. Is composed of sulfated glycoproteins. b. Prevents polyspermy c. Prevents premature implantation.
d. Functions as a porous filter e. Arises by delamination from the embryoblast during the 2nd week.

_____4. Before a **spermatozoan fertilizes an ovum**:

- a) The zona pellucida must be shed. b) Its glycoprotein coat must be removed from its head.
c) Its acrosomal membrane must be impermeable. d) Its tail piece should be lost.
e) It should be capacitated in the epididymidis.

_____5. With regard to **fertilization**, which statement is **false**? **Fertilization**:

- a) Results in variation of human species through mingling of maternal and paternal chromosomes.
b) Causes metabolic activation of the ovum and initiates cleavage of the zygote.
c) Restores the normal diploid number of chromosomes in the zygote.
d) Determines chromosomal sex of the embryo. e) None of the above.

_____6. During the **first week of development**,

- a) The prochordal plate marks the site of the future mouth.
b) Developmental disturbances may give rise to major congenital anomalies.
c) The blastocyst is superficially implanted in the uterine wall (endometrium).

- d) The amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.
 e) The embryo is a solid ball of blastomeres and is called gastrula, a process which establishes the three definitive germ layers.

_____ **7. During the second week of development, which statement is true?**

- a) Epiblast and hypoblast are the two layers of extraembryonic mesoderm.
 b) In placenta previa, high levels of hCG are found but no inner cell mass (embryo) is present.
 c) Amniocentesis will show high levels of alpha-fetoprotein with neural tube or ventral wall defects.
 d) The cells of the embryoblast form the hypoblast and epiblast, which begin to form the amniotic cavity.
 e) None of the above.

_____ **8. Which of the following is/are not a derivative(s) of the neural crest?**

- a) Schwann cells b) chromaffin cells c) neurohypophysis d) melanocytes e) parafollicular (C) cells

_____ **9. What germ layer is the dermis derived from?**

- a) Paraxial mesoderm b) Intermediate mesoderm c) Lateral plate mesoderm
 d) Neural crest e) Endoderm

_____ **10. Which of the following is/are not mesodermal in origin?**

- a) microglia b) bone and cartilage c) astrocytes
 d) cardiac muscle cells e) endothelium of blood and lymph vessels

_____ **11. Intermediate mesoderm gives rise to the**

- a) neural tube b) kidneys and gonads c) heart d) somites e) notochord

_____ **12. During the third week of development, which statement is false?**

- a) Somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
 b) The primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
 c) The notochord, which induces the overlying ectoderm to develop into the neural plate, forms the primitive axis of the embryo around which the axial skeleton will form.
 d) Mesoderm exists between the ectoderm and endoderm everywhere. e) Neural tube is formed.

_____ **13. During the fetal period, which statement is true?**

- a) The spleen is the major site of erythropoiesis during the end of the 9th week.
 b) The bodies of 20-week fetuses are usually completely covered with lanugo hairs.
 c) By 18 weeks the uterus is formed and canalization of the vagina has begun.
 d) Pupillary light reflex occurs during weeks 26-29. e) the amount of white fat is about 16% of body weight at 29 weeks.

_____ **14. With regard to the placenta, which statement is true?**

- a) Deoxygenation of blood is one of the major functions of the placenta.
 b) As pregnancy advances, the placental membrane becomes progressively thicker.
 c) The fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
 d) The maternal part of the placenta develops from the chorionic sac. e) Amniochorionic membrane ruptures during labor.

_____ **15. Concerning the fetal membranes (developmental adnexa), which statement is true?**

- a) The yolk sac helps control the embryo's body temperature.
 b) As the bladder enlarges, the umbilical cord involutes to form urachus.
 c) Blood formation occurs in the wall of allantois during the third to fifth weeks.
 d) Primordial germ cells appear in the endodermal lining of the wall of the allantois.
 e) During the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.

_____ **16. Concerning multiple pregnancies,**

- a) Twins with one amnion, one chorion and one placenta are always monozygotic.
- c) The frequency of monozygotic twinning shows marked racial differences.
- b) The rate of dizygotic twinning increases with maternal age.
- d) Dizygotic twins always have two amnions and two chorions. e) monozygotic twins are of the same sex.

_____ **17. Sacroccygeal teratoma is a tumor that arises from remnants of the**

- a) neural plate b) primitive streak c) posterior neuropore d) cloacal membrane e) notochord

_____ **18. Most birth defects occur during:**

- a) The first 2 weeks b) The embryonic period c) The fetal period
- d) Early childhood e) None of the above

_____ **19. With regard to birth defects, which statement is **false**?**

- a) Clubfoot, which is due to mechanical forces in the womb, is an example of deformation.
- b) Trisomy of the autosomes occurs with increasing frequency as maternal age increases.
- c) Deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
- d) The phenotype of Turner syndrome is female. e) Thalidomide induces limb defects.

_____ **20. Failure of fusion of the neural arches of the developing vertebrae could lead to:**

- a) Achondroplasia b) Acromegally c) Brachycephaly d) Spina bifida e) Amelia

_____ **21. With regard to the development of the musculoskeletal system,**

- a) The muscular system develops from neuroectoderm.
- b) The upper limbs rotate laterally through 90° on their longitudinal axes.
- c) Premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) Sclerotomes of the somites form the viscerocranium. e) Xiphoid process develops during the 6th week.

_____ **22. With regard to the pharyngeal apparatus, which statement is **false**?**

- a) The fourth inferior pouch forms the ultimopharyngeal body.
- b) The second aortic arch forms the hyoid and stapedial arteries.
- c) The second pharyngeal/branchial arch is associated with muscles of facial expression.
- d) The third superior pouch forms the inferior parathyroid _____ e) None of the above.

_____ **23. Structure that is **not** developed in the third branchial arch include:**

- a) Lesser cornu of hyoid bone b) Body of the hyoid bone c) Glossopharyngeal nerve
- d) Common carotid artery e) Stylopharyngeus muscle

_____ **24. Which of the following structures develop from endoderm of the second pharyngeal pouch?**

- a) Thymus gland b) Palatine tonsil c) Thyroid gland
- d) Superior parathyroid gland e) Parafoallicular cells of thyroid gland

_____ **25. With regard to the development of the head and neck, which statement is **false**?**

- a) Fontanelle is a single (not paired) bone derived from the viscerocranium.
- b) Median cleft of the lower lip involves failure of the mandibular prominence to fuse.
- c) The primary palate forms the premaxillary part of the maxilla and holds the incisors.
- d) All of the flat bones on top of the skull are formed from the membranous neurocranium.
- e) Excessive merging of the maxillary and mandibular prominences would cause microstomia.

- _____ **26.** Which of the following **muscles** develop from the **epimeres**?
- a) Intercostal muscles. b) The strap muscles of the neck. c) Rectus abdominis.
d) Quadratus lumborum. e) Extensor muscles of the back.
- _____ **27.** All of the following **appendages of the skin** develop from ectoderm, **except**:
- a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.
- _____ **28.** Concerning the development of the **integumentary system**, which statement is **false**?
- a) Epidermis is derived from surface ectoderm.
b) Teeth develop from ectoderm, mesoderm and neural crest cells.
c) Hair follicles begin to proliferate from the stratum corneum of the epidermis.
d) Toenails and fingernails begin to develop at the tips of the digits postnatally.
e) Castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- _____ **29.** Concerning the development of the **circulatory system**, which statement is **false**?
- a) All arteries, veins, and lymphatic channels form from mesoderm
b) The transverse pericardial sinus forms from the breaking away of dorsal mesentery.
c) The proximal third of the bulbus cordis becomes the right ventricle.
d) The middle third (conus cordis) of the bulbus cordis becomes the left and right atria.
e) Vagus nerve is associated with the aortic arches IV and VI.
- _____ **30.** What major structure is formed by the **sinus venosus**?
- a) Left Atrium b) Right Atrium c) Aorta d) Left Ventricle e) Right Ventricle
- _____ **31.** Which of the following is **NOT associated with Tetralogy of Fallot**?
- a) Pulmonary stenosis b) Pulmonary atresia c) Ventricular septal defect
d) Right ventricular hypertrophy e) Dextroposition of the Aorta (over riding aorta)
- _____ **32.** During fetal life, in which of the following structures is **the percent hemoglobin/oxygen saturation level** of fetal blood the lowest?
- a) Right Atrium b) Inferior Vena Cava c) Umbilical Vein d) Ductus Arteriosus e) Descending Aorta
- _____ **33.** The **common carotid arteries** develop from:
- a) Truncus arteriosus b) 1st aortic arch arteries c) 2nd aortic arch arteries
d) 3rd aortic arch arteries e) Dorsal aortae
- _____ **34.** Which of the following veins participates in the **formation of the inferior vena cava**?
- a) Right vitelline vein b) Right posterior cardinal vein
c) Right umbilical vein d) Left umbilical vein e) Left posterior cardinal vein
- _____ **35.** **Portal vein** is a derivative of:
- a) Anastomosis between right umbilical and right vitelline veins
b) Anastomosis between left umbilical and left vitelline veins
c) Right vitelline vein d) Left vitelline vein e) Anastomosis between left and right vitelline veins
- _____ **36.** Concerning the **development of vasculature**,
- a) Distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
b) The left 4th aortic arch becomes the proximal part of the left subclavian artery.

- c) The distal part of the right 6th aortic arch becomes the ductus arteriosus.
- d) The superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) After birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.

_____ **37. Which of the following *cells of the lung* develop from mesoderm?**

- a) Type I Pneumocytes.
- b) Type II Pneumocytes.
- c) Alveolar dust cells.
- d) Clara cells
- e) Goblet cells

_____ **38. Which of the following *salivary glands* develop from ectoderm?**

- a) Parotid
- b) Submandibular
- c) Sublingual
- d) Lingual
- e) All of the above

_____ **39. With regard to the *development of the digestive system*, which statement is *false*?**

- a) the foregut may give rise to the lower respiratory system.
- b) the hindgut derivatives are supplied by inferior mesenteric artery.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.
- e) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.

_____ **40. Concerning the *development of the GIT and associated structures*, which statement is *true*?**

- a) Stomach rotates 270° counterclockwise, similar to the midgut.
- b) The anal membrane normally breaks down by the end of the second trimester.
- c) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- d) Meckel's diverticulum is a congenital connection from the umbilicus via a vitelline ligament to the ileum.
- e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from local mesoderm.

_____ **41. The *crura of the diaphragm* develops from:**

- a) Septum transversum.
- b) Dorsal mesentery of the esophagus.
- c) Pleuroperitoneal membrane.
- d) Myotomes of the body wall.
- e) None of the above.

_____ **42. Concerning the *development of the urinary system*, which statement is *false*?**

- a) Urinary system begins to develop before the genital system
- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **43. Concerning the *development of the urinary bladder and urethra*, which statement is *false*?**

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) The urinary bladder becomes a pelvic organ during the 9th week of development.
- c) In adults in both sexes, the urachus is represented by the median umbilical ligament.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ **44. With regard to the *congenital anomalies of the kidneys and ureters*, which statement is *false*?**

- a) Accessory renal arteries are end arteries.
- b) Bilateral renal agenesis is associated with oligohydramnios.
- c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.

d) Unilateral renal agenesis is more common in males than females. e) None of the above.

_____ **45.** During the development of the urinary system, which statement is true?

- a) Connecting tubule is the adult derivative of the ureteric bud
- b) The podocytes of Bowman's capsule are derived from endoderm and ectoderm.
- c) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- d) Urachal cyst is a remnant of the cloaca. e) Urine production begins shortly after birth.

_____ **46.** With regard to the development of the endocrine system, which of the following statements is false?

- a) The pituitary gland is ectodermal in origin.
- b) The thyroid gland is the first endocrine gland to develop in the embryo.
- c) The pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.
- d) C cells of the thyroid gland differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
- e) The oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.

_____ **47.** With regard to the development of gonads, which statement is false?

- a) Sex of an embryo is determined genetically at the time of fertilization.
- b) Germ cells do not normally appear in the genital ridges until the 6th week of development.
- c) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.
- d) The surface epithelium of the female gonad develop into oogonia. e) None of the above.

_____ **48.** During the development of the female reproductive system, which statement is false?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis. e) No oogonia are formed postnatally.

_____ **49.** During the development of the male reproductive system, which statement is false?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- c) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- d) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
- e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the mesonephric ducts.

_____ **50.** During the development of the reproductive system, which statement is true?

- a) Genital tubercle during weeks 3-6 becomes the clitoris.
- b) The lower portion of the vagina is derived from sinovaginal bulb.
- c) In the male, failure of urethral folds to fuse completely results in cryptorchidism.
- d) Dihydrotestosterone inhibits the growth of the paramesonephric ducts in males.
- e) In Turner's syndrome the vagina ends in a blind pouch and testes are lodged in the inguinal canal.

_____ **51.** With regard to the vestigial structures derived from embryonic genital system, which statement is false?

- a) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
- b) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- c) A structure found within the adult female pelvis formed from the gubernaculum is median umbilical ligament.

d) The cranial end of mesonephric duct in males may persist as paradidymis. _____ e) None of the above.

_____ **52.** Concerning the **development of the genital system**, one statement is **false?**

- a) Gubernaculum does not play any role during the descent of ovaries.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.
- d) Inguinal canals develop in both sexes. _____ e) None of the above.

_____ **53.** **Primary brain vesicle** that remains relatively unchanged as compared to the others is:

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalon

_____ **54.** **Cerebellum** develops from cells of:

- a) Basal plate of myelencephalon
- b) Alar plate of myelencephalon
- c) Basal plate of metencephalon
- d) Alar plate of myelencephalon
- e) Basal plate of nesencephalon

_____ **55.** During the **development of the central nervous system** zone of the wall of the neural tube to be occupied by neuroblasts forming the gray matter is:

- a) Marginal zone
- b) Ventricular zone
- c) Mantle zone
- d) Ependymal zone
- e) Neuroepithelial zone

_____ **56.** The **neuroepithelial cells of the neural tube** gives rise to all of the following cells of the brain except:

- a) Microglia.
- b) Astrocytes.
- c) Oligodendrocytes.
- d) Purkinje cells of cerebellar cortex.
- e) Ependymal cells.

_____ **57.** Which of the following structures in the **eye** is derived from **surface ectoderm?**

- a) retina
- b) dilator and sphincter pupillae muscles of iris
- c) corneal endothelium
- d) vitreous body
- e) lens

_____ **58.** Which of the following statements about the **optic nerve** is **false?** The **optic nerve**

- a. is a tract of diencephalon.
- b. is myelinated by oligodendrocyte.
- c. is invested by meninges.
- d. is not capable of regeneration after transaction.
- e. is developed from the first pharyngeal pouch.

_____ **59.** With regard to **development of the eye**, which statement is **true?**

- a. The hyaloid canal is found in the vitreous body.
- b. Aqueous humor is produced by the choroid plexus.
- c. Failure of choroid fissure to close results in congenital aphakia.
- d. Sclera is continuous with the dura of the central nervous system.
- e. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes congenital cataract.

_____ **60.** With regard to **development of the ear**, which statement is **true?**

- a. The middle ear cavity develops from pharyngeal pouch 1.
- b. Perilymph enters the subarachnoid space via the ductus reuniens.
- c. Otic placode and neural crest form the vestibulocochlear ganglion.
- d. The saccular portion of the otic vesicle gives rise to the endolymphatic duct.
- e. Utricular portion of the otic vesicle, which is endodermal in origin, gives rise to the organ of Corti.

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